

EXPERIMENT NO: 02

TITLE: Cross-Platform Prompting — Evaluating Diverse Techniques in AI-Powered Text Summarization

AIM:

To evaluate and compare the effectiveness of different prompting techniques — **Zero-shot**, **Few-shot**, **Chain-of-Thought**, and **Role-based prompting** — across multiple AI platforms such as **ChatGPT**, **Gemini**, **Claude**, and **Copilot**, in performing the task of text summarization for a technical article titled *“The Basics of Blockchain Technology.”*

SCENARIO:

You are part of a **content curation team** for an educational platform that creates concise and accurate summaries of research papers for undergraduate students.

The given task is to summarize a **500-word article** on *“The Basics of Blockchain Technology”* using four AI platforms and four prompting techniques, in order to identify which combination provides the **best performance** in terms of:

- **Accuracy** (correctness of information)
 - **Coherence** (logical flow and structure)
 - **Simplicity** (ease of understanding)
 - **Speed** (response time)
 - **User experience** (clarity, readability, and satisfaction)
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ALGORITHM / PROCEDURE:

1. Input Selection:

- Choose a 500-word technical article on *“The Basics of Blockchain Technology.”*

2. Prompt Design:

- Create four different prompts for the same summarization task using different techniques:

- **Zero-shot Prompt:**

“Summarize the following article on the basics of blockchain technology in 100 words.”

- **Few-shot Prompt:**

“Here is an example of a short summary of a technical topic: [example].

Now summarize the article on blockchain technology in a similar style, within 100 words.”

- **Chain-of-Thought Prompt:**

“Explain step-by-step what blockchain is, how it works, and then provide a short 100-word summary of the article.”

- **Role-based Prompt:**

“You are an expert educator creating simple summaries for undergraduate students. Write a 100-word summary of the article on blockchain technology, focusing on clarity and accuracy.”

3. Platform Execution:

- Run each prompt on four AI platforms:
 - ChatGPT
 - Google Gemini
 - Anthropic Claude
 - Microsoft Copilot

4. Data Recording:

- Record the output summaries from each platform and note the following parameters:
 - Time taken to generate the summary.
 - Quality of the output (in terms of accuracy, coherence, simplicity).

5. Evaluation Criteria:

- Use a 5-point rating scale (1 = Poor, 5 = Excellent) for each factor:

Technique / Platform	Accuracy	Coherence	Simplicity	Speed	User Experience
Zero-shot (ChatGPT)	4	4	4	5	5
Few-shot (ChatGPT)	5	5	4	4	5
Chain-of-Thought (ChatGPT)	5	5	5	4	5
Role-based (ChatGPT)	5	5	5	5	5
Zero-shot (Gemini)	4	4	3	5	4
Few-shot (Claude)	5	5	4	4	5
Chain-of-Thought (Claude)	5	5	4	4	4
Role-based (Copilot)	4	4	4	5	4

6. Comparison and Analysis:

- Compare the results across all platforms and prompting styles.
- Identify which combination provides the best balance of accuracy, coherence, and simplicity for student-level summaries.

RESULT:

After performing the experiment, it was observed that:

- **ChatGPT with Role-Based Prompting** and **ChatGPT with Chain-of-Thought Prompting** produced the most accurate, coherent, and easy-to-understand summaries.
- **Gemini** generated fast responses but was less detailed.
- **Claude** maintained strong coherence but required clearer instructions.
- **Copilot** was quick but less creative in phrasing.

Final Evaluation:

Platform	Best Prompt Technique	Overall Score (/25)	Remarks
ChatGPT	Role-based Prompting	25	Most balanced and effective
Claude	Few-shot Prompting	22	Good coherence and clarity
Gemini	Zero-shot Prompting	20	Fast but less detailed
Copilot	Role-based Prompting	21	Good for concise summaries

Conclusion:

Among all evaluated models and prompting techniques, **ChatGPT with Role-based or Chain-of-Thought prompting** gave the **best summarization results** in terms of **accuracy, coherence, simplicity, and user experience**.

It demonstrated that carefully designed prompts significantly enhance output quality, making prompt engineering a vital skill for effective use of AI platforms in educational content curation.
